

KHAMAR-DABAN, Russian Federation

WMO: 308150

Lat: **51.5297N**Lon: **103.5890E**Elev: **1442**StdP: **85.16**Time Zone: **8.00 (E08)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -27.2	(d) -30.7	(e) 0.3	(f) -29.4	(g) -28.6	(h) 0.3	(i) -26.9	(j) 6.8	(k) -12.2	(l) 5.0	(m) -12.4	(n) 0.9	(o) 30	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.9	(c) 23.1	(d) 14.4	(e) 21.4	(f) 13.9	(g) 19.7	(h) 13.1	(i) 16.1	(j) 20.1	(k) 15.1	(l) 19.3	(m) 14.2	(n) 18.0	(o) 2.5	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.3	17.7	13.6	11.6	16.5	12.7	10.9	15.6	50.4	20.2	47.3	19.5	44.8	18.1	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.2	(b) 4.2	(c) 3.4	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 5.2	(b) 4.2	(c) 3.4	(d) WB	(e) -32.1	(f) 17.9	(g) 3.4	(h) 1.1	(i) -34.6	(j) 18.7	(k) -36.5	(l) 19.4	(m) -38.5	(n) 20.0	(o) -40.9	(p) 20.8	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-15.7	-14.8	-8.2	-1.8	4.3	11.2	13.5	12.1	5.9	-1.8	-10.2	-13.3
	DBStd	11.70	5.75	6.09	6.47	5.34	5.13	4.26	2.92	3.90	5.32	5.55	6.86	5.62
	HDD10.0	4502	797	695	565	355	189	37	4	21	144	366	606	723
	HDD18.3	7246	1056	928	823	603	436	217	153	196	372	623	856	982
	CDD10.0	306	0	0	0	2	12	71	112	87	22	1	0	0
	CDD18.3	9	0	0	0	0	1	2	2	3	1	0	0	0
	CDH23.3	42	0	0	0	0	3	11	8	15	4	0	0	0
	CDH26.7	2	0	0	0	0	0	1	0	1	0	0	0	0
	WSAvg	1.4	1.2	1.2	1.4	1.5	1.5	1.6	1.5	1.5	1.6	1.4	1.2	1.2
	PrecAvg	1401	31	29	56	96	112	185	304	230	159	83	62	43
(15) Precipitation	PrecMax	1986	53	56	95	183	222	500	726	521	408	181	111	100
	PrecMin	1102	10	6	26	46	30	46	97	91	66	32	19	0
	PrecStd	211	13	13	17	29	37	91	144	109	64	34	22	18
	PrecStd	211	13	13	17	29	37	91	144	109	64	34	22	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.6	1.3	9.8	16.2	22.5	24.9	24.4	25.2	23.2	15.0	8.1	1.9
		MCWB	-4.5	-3.1	2.5	5.8	10.6	15.0	16.7	15.4	12.0	5.6	1.5	-2.2
	2%	DB	-3.3	-1.2	6.9	12.1	18.4	22.4	22.5	22.6	19.1	11.0	4.7	-1.3
		MCWB	-6.4	-4.9	0.7	3.9	8.8	13.4	15.3	14.6	10.2	3.8	-0.1	-4.7
	5%	DB	-5.7	-3.6	4.3	9.2	15.6	20.3	21.0	20.4	16.3	8.6	1.7	-4.0
		MCWB	-8.0	-6.5	-0.9	2.3	7.5	12.5	14.9	13.9	10.0	2.4	-2.1	-6.2
	10%	DB	-8.0	-6.0	1.7	6.9	13.3	18.3	19.3	18.6	14.3	6.5	-1.0	-6.0
		MCWB	-9.8	-8.4	-2.6	1.3	6.3	11.7	14.4	13.2	8.8	1.1	-3.9	-7.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.5	-3.0	2.8	6.6	11.7	16.2	18.1	16.9	13.5	5.9	1.8	-2.0
		MCDB	-1.0	0.7	9.1	14.6	21.1	21.5	23.3	22.9	18.1	13.8	7.2	1.6
	2%	WB	-6.3	-4.8	0.8	4.4	9.5	14.6	16.6	15.5	11.9	4.3	0.0	-4.4
		MCDB	-3.6	-1.5	6.6	11.2	17.2	19.8	20.1	20.2	16.8	10.5	4.4	-1.8
	5%	WB	-7.9	-6.4	-0.8	2.8	8.1	13.6	15.6	14.6	10.5	2.7	-1.8	-6.0
		MCDB	-5.9	-3.7	4.0	8.5	14.6	18.1	19.2	19.0	15.6	7.8	1.5	-4.0
	10%	WB	-9.7	-8.3	-2.5	1.4	6.5	12.7	14.6	13.7	9.2	1.5	-3.7	-7.7
		MCDB	-8.2	-6.0	1.4	6.6	12.7	16.7	18.2	17.6	13.9	5.8	-1.0	-6.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.2	10.0	9.5	9.8	8.5	7.9	7.5	8.5	8.5	8.8	8.2
		MCDBR	10.9	10.6	12.5	13.0	13.6	10.9	10.5	10.3	10.8	11.5	11.0	10.0
		MCWBR	8.1	7.8	7.5	7.0	6.9	5.5	5.8	5.7	5.6	6.5	7.0	7.4
		MCDBR	11.1	10.7	12.2	11.9	12.6	8.9	8.8	8.9	9.9	10.8	10.6	9.5
	5% WB	MCWBR	8.4	7.9	7.5	6.6	6.7	5.2	5.5	5.3	5.6	6.5	6.9	7.0
(40) Clear-Sky Solar Irradiance	taub		0.233	0.257	0.295	0.337	0.381	0.373	0.369	0.342	0.298	0.280	0.254	0.235
	taud		2.232	2.213	2.205	2.299	2.218	2.355	2.401	2.461	2.546	2.498	2.310	2.190
	Ebn at Noon		801	875	901	901	871	881	878	888	901	851	765	723
	Edh at Noon		66	90	112	117	133	117	110	99	80	68	61	58
(44) All-Sky Solar Radiation	RadAvg		1.39	2.47	3.88	4.94	5.42	5.47	5.09	4.44	3.57	2.34	1.41	1.00
	RadStd		0.05	0.10	0.18	0.25	0.26	0.35	0.38	0.29	0.31	0.14	0.08	0.05

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page